EUDR Schema and System Design

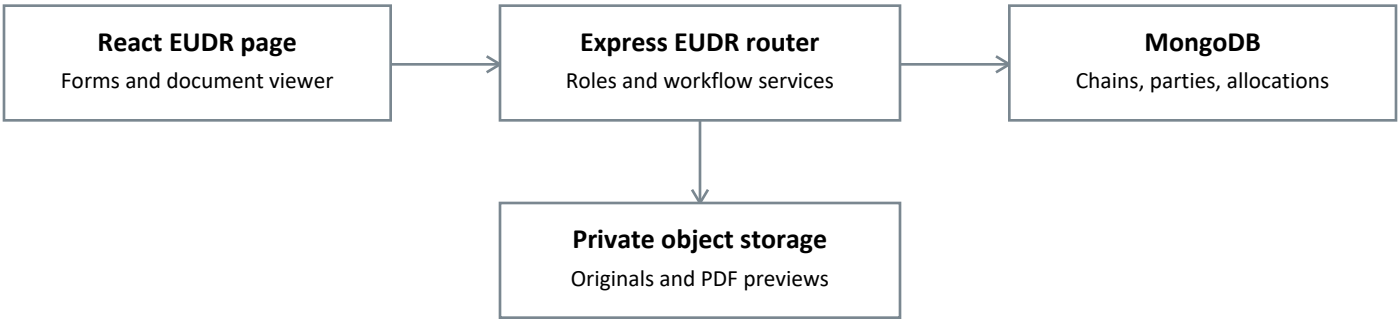
Green House Sourcing OMS

Technical guide | 13 October 2026 | Code snapshot 9999690

This document explains how our EUDR timber traceability module stores data, moves a chain through verification, reconciles timber volumes and protects supporting documents. It is intended for developers, administrators and people maintaining the workflow.

The central record is a chain for one container and shipping vendor. Several chains may contribute to the same vendor's stuffed timber total. Each chain embeds its ordered stages, evidence metadata and audit history; document bytes are stored separately.

1 Architecture at a glance



Layer	Responsibility
React and Vite	Container lookup, contributions, stage forms, saved-party selection, review and PDF/image rendering.
Express controllers	Authorize requests, load live OMS data, save and verify stages, stream evidence and manage errors.
Domain services	CFT reconciliation, workflow rules, invoice keys, allocation transactions, directory snapshots and previews.
MongoDB and Wasabi	MongoDB stores structured records; S3-compatible Wasabi storage holds originals and derived previews.

2 Domain relationships

Existing Orders and Samples supply shipment rows. Orders resolve an Item by item code; samples carry their own timber_cft. Each row resolves to one Vendor. EudrChain references that vendor and embeds Stage records. A receiver links back to a seller stage through incoming_invoice_stage_id.

There is no separate EUDR container or invoice collection. Container/vendor grouping, stuffed totals and coverage are derived. EudrParty is a reusable directory, not the owner of historical chain identities. EudrAllocation is a concurrency lock, not an inventory ledger.

Reading guide: chain schema on page 2; embedded and directory schemas on page 3; workflow on page 4; consistency on page 5; APIs on page 6; files and security on page 7; maintenance on page 8.

Chain schema and OMS connections

Collection: **eudr_chains** | Mongoose model: **EudrChain**. One document represents one chain. The schema uses timestamps and optimistic concurrency through the version key.

Field	Stored type and meaning
_id	ObjectId. Chain identifier generated by MongoDB/Mongoose.
container container_key	Strings, required. Display container and its trimmed lowercase lookup key.
vendor	Required ObjectId reference to Vendor. The shipping vendor for this chain.
declared_milli_cft	Number, required. Container/vendor declared total in thousandths of CFT. API requires a positive value with at most three decimals.
contribution_milli_cft	Number or null. This chain's allocated contribution, at least 1 milli-CFT. Legacy null falls back to declared_milli_cft.
stages	Ordered array of embedded stages. New chains start with a sawmill stage at workflow_version 2.
invoice_keys	Array of normalized invoice identity strings. Omitted when empty; backs the sparse unique index.
history	Embedded audit events, including edits, verification, invalidation, replaced evidence and superseded stages.
created_by createdAt and updatedAt	Creator ObjectId referencing users, plus automatic creation and update dates.
__v	Mongoose version key. Concurrent edits fail with a conflict instead of silently overwriting a chain.

Indexes and uniqueness

A non-unique index on (container_key, vendor, createdAt) supports multiple chains per vendor. A unique sparse multikey index on invoice_keys prevents two active chain documents from registering the same invoice key. Empty invoice arrays are unset before validation so multiple drafts can exist.

Existing OMS fields

Model field	Use in EUDR
Vendor.gst and Vendor.pan	Trimmed uppercase strings, empty by default. Both are required to start or verify a chain.
Item.timber_cft	Optional non-negative Number, default null. Per-unit CFT maintained through the PIS update.
Sample.timber_cft	Optional non-negative Number, default null. Used with sample shipment quantities.

Stuffed CFT, shipment rows, excluded rows, coverage, current_stage_id and complete are response calculations, not persistent chain columns.

Embedded records and reusable directory

Stage schema

Field group	Type and purpose
_id and kind	Automatic ObjectId; kind is sawmill, transport, receiver or declaration.
workflow_version	Number. Schema default 1 supports legacy stages; new and edited stages use version 2.
fields	Mixed object with a stage-specific allowlist. Saves normalize values to trimmed strings, including CFT and coordinates. Completeness is enforced by the service before verification.
incoming_invoice_stage_id	String containing the preceding seller stage ID, or empty for unlinked legacy stages. Receiver identity and incoming evidence derive from that seller.
documents	Embedded evidence metadata array. The upload route keeps one current attachment per document type; replacing it preserves the old metadata in history.
verified_at and verified_by verified_by_name	Nullable Date and users ObjectId, plus reviewer name snapshot. verified_at drives workflow gating.
verification_method verification_result	Method string defaults to manual. Result enum is pending or verified; there is no rejected-state workflow or AI implementation.
vendor_signature shipment_signature	SHA-256 signatures used to detect changed vendor identifiers or shipment/CFT inputs at later reads and writes.

Document and audit records

Each document has an automatic **_id**, required **type**, storage **key** and original **name**, plus content_type, size, sha256 and uploaded_at. File bytes are not stored in MongoDB. A history entry stores event, stage_id, at, actor, actor_name and a Mixed detail object containing the relevant values or snapshots.

EudrParty and EudrAllocation

eudr_parties stores kind (sawmill, transporter, vehicle or buyer), identity_key, name, optional parent_key, Mixed fields, evidence metadata and timestamps. Its unique index is (kind, identity_key); (kind, parent_key, name) supports directory lookup. Vehicle entries combine truck and driver details rather than using a separate driver collection.

Sawmill and buyer keys combine normalized GST and PAN. A transporter key uses GST when present, otherwise its normalized name. A vehicle key adds truck and driver-licence identifiers to the transporter key. Only reusable identity and approved evidence categories are saved; transaction numbers, quantities and dates are excluded.

eudr_allocations has a String _id formed from container_key and vendor ID, and a revision Number. Incrementing this record inside a transaction serializes allocation changes for that container/vendor.

Workflow and required stage data

Sawmill - Transport - Receiver - repeat Transport and Receiver if needed - Declaration

Stages are created by verification, not supplied as arbitrary client-defined sequences. A later stage cannot be saved, uploaded to or verified until its immediate predecessor is verified. Authorized users may verify their own entries.

1 Sawmill and origin

Fields: name, gst, pan, permission_number, forest_latitude, forest_longitude, invoice_number, invoice_date, sold_to, buyer_gst, buyer_pan and sold_cft. Optional: cutting_limit_cft. Buyer selection also uses buyer_type (vendor or other) and buyer_vendor_id when a vendor is selected.

Evidence types: cutting_invoice, license, permission, forest_image, buyer_gst and buyer_pan. The UI calls cutting_invoice the Sawmill Sale Invoice; the internal key remains unchanged for compatibility. Latitude must be within -90 to 90 and longitude within -180 to 180. Forest evidence must be an actual PNG, JPEG or WebP image.

2 Transport

Fields: transporter_name, truck_number, driver_license_number, ewaybill_number and billty_number. Optional: gst and truck_registration_number. **Evidence:** weight_slip, billty, driver_license and ewaybill; truck_registration becomes mandatory when its registration number is entered.

3 Receiver and onward sale

New receivers inherit name, gst and pan from the preceding seller's buyer. incoming_invoice_stage_id references that seller's invoice and file without a new upload or invoice registration. received_cft starts blank and must be entered explicitly; it is non-negative with at most three decimals. Receipt differences do not block verification.

Both normalized GST and PAN must match the shipping vendor. A matching receiver has no onward sale or new uploads and opens declaration after verification. Otherwise it must record sold_to, buyer_gst, buyer_pan, sold_cft, invoice_number and invoice_date, plus invoice, buyer_gst and buyer_pan evidence. Its own identity is the outgoing invoice issuer; verification opens another transport stage.

4 Declaration

Requires declaration evidence and final manual verification for each chain. Final checks include exact declared-versus-stuffed CFT, a valid final receiver match, a contribution within capacity, and resolution of unassigned shipment rows. A verified chain can coexist with incomplete overall vendor coverage.

Rules shared by all sellers

Sold CFT must be positive, have at most three decimals and cover the chain contribution; it may exceed that contribution. Vendor selection loads buyer identity server-side. Other-party details are saved for reuse. Correcting receiver identity requires editing the preceding seller, which supersedes downstream stages.

CFT reconciliation and consistency controls

How stuffed timber is calculated

Container lookup reads positive shipment quantities from Orders and Samples. Order items resolve by item code; samples use their own per-unit timber_cft. Calculations use integer thousandths of a cubic foot, called milli-CFT. Each shipment line is rounded before totals are summed, avoiding equality checks on raw decimal totals.

For each line: convert per-unit CFT to milli-CFT, multiply by shipped quantity, then round. Sum included line amounts for each shipping vendor and divide by 1,000 for display. GST, PAN and invoice numbers normalize to uppercase alphanumeric characters; container keys use trimmed lowercase text.

Example	CFT	Interpretation
Stuffed and declared	17.600	The overall container/vendor requirement.
Chain A contribution	10.000	Counts once when this chain is currently verified.
Chain B contribution	7.600	Completes coverage with Chain A, even if invoices sell larger quantities.

coverage.complete requires a positive stuffed total, allocated contributions exactly equal to it, every allocated contribution currently verified, and no unassigned shipments. Missing timber_cft rows are excluded with a warning; a numeric zero is not a missing value. Ambiguous vendors, including multi-vendor samples, appear as unassigned and block final declaration.

Atomic allocation and invoice identity

Create, contribution edits and declared-total updates use a MongoDB transaction and the shared allocation revision. The lock is taken before reading sibling chains, preventing simultaneous requests from allocating the same remaining capacity. Standalone MongoDB without transaction support returns 503 and saves no allocation changes. Declared-total edits synchronize sibling chain totals without inflating explicit contributions.

Invoice key format is normalized issuer GST | financial-year start year | normalized invoice number. The Indian year changes on 1 April. For example, 31 March 2026 belongs to FY 2025-26, while 1 April 2026 belongs to FY 2026-27. Uniqueness is global across active chain invoice_keys; linked incoming invoices do not register again.

When verification becomes stale

Change	Effect
Stage fields or evidence	Clear this stage’s approval; archive later stages in history and remove them from the active sequence.
Chain contribution	Clear all approvals in this chain; keep stages, evidence and audit history.
Vendor GST or PAN	Invalidate the final receiver and downstream approval when signatures differ.
Shipment contents or item CFT	Invalidate declaration approval when the shipment signature differs.

Invalidation is evaluated lazily when chains are read or changed, not by a background event subscriber. GET list/detail can therefore persist invalidation and audit entries. Shipment data is recalculated rather than frozen at chain creation.

EUDR API contract

Backend mount: **/eudr**. Every route requires authentication and an admin, super_admin or manager role. Responses are private and no-store. The paths below are relative to this mount; any deployment API prefix is outside this router.

Method and path	Purpose and main inputs
GET /container	Query container. Returns vendors, shipment rows, exclusions, unassigned rows, chain summaries and coverage.
GET /	Optional container query. Returns up to 50 recent chains, newest updated first.
POST /	Create with container, vendor_id, declared_cft and optional contribution_cft. Omitting contribution uses the declared total.
GET /:id	Return full chain, live stuffed CFT, coverage, current stage and completion flags.
PATCH /:id/declared	Body declared_cft. Must equal live stuffed CFT; updates the container/vendor's sibling totals.
PATCH /:id/contribution	Body contribution_cft. Positive, three-decimal contribution within remaining capacity.
GET /directory	Query kind, optional search and parent_key. Kinds include vendor plus saved-party kinds. Maximum 50 matches.
GET /directory/:partyId	Return saved identity and evidence identifiers; evidence storage keys are stripped.

For stage operations, **S = /:id/stages/:stageId**. For document operations, **D = S/documents/:documentId**.

Method and path	Purpose and main inputs
PUT S	Body fields object. Replaces the field snapshot; does not merge a partial object. Unknown fields and values over 500 characters are rejected. Incomplete drafts may be saved.
POST S/documents	Multipart file plus type. One file per request and one current file per evidence type.
POST S/reuse-evidence	Body party_id, evidence_id and confirmed: true. Server validates matching identity and duplicate restrictions.
POST S/verify	Validate completeness and preceding approval, record manual verification and create the next stage when needed.
GET D/url	Issue a 300-second signed original URL; download=1 requests original download.
GET D/content	Stream original content through the API with HTTP Range support.
GET D/preview	Return the original image/PDF or a private converted Office PDF.

Response and error conventions

Chain mutations and detail use {data: chain}; lists and directories use {data: [...]}; URL returns {url}; container lookup returns grouping fields directly. Common errors include 400 validation, 403 forbidden, 404 missing record, 409 duplicates/concurrent edits/gating/CFT mismatch, and 503 unavailable transactions or conversion capacity. Preview conversion also uses 413 for size and 422 for unconvertible output.

Documents and security design

Upload and reuse lifecycle

Uploads use a dedicated EUDR extension and MIME allowlist, not the generic upload policy. Supported formats are PDF, PNG, JPEG, WebP, DOC, DOCX, XLS, XLSX, CSV and TXT. Default upload limit is 50 MiB, configurable through GENERIC_UPLOAD_MAX_FILE_SIZE. Forest images additionally pass actual-byte decoding and format checks.

The server checks normalized filename and SHA-256 content against current and archived chain evidence. A renamed copy still conflicts, and the same filename conflicts even with different bytes. It uploads to a generated key under eudr/{chainId}/{stageId}, then saves metadata and invalidates affected approvals. A newly uploaded object is deleted if the chain save fails.

Multiple evidence types are supported within a stage, but there is one current attachment per type. Replacement moves the old metadata into history. All Documents contains only current stage attachments, so linked invoices and audit copies are not counted twice.

Saved evidence reuse accepts directory identifiers, never an arbitrary storage key. It requires explicit validity confirmation, matches the stage's sawmill or truck/driver identity, and repeats destination-chain filename/hash checks. Reuse references the existing private object rather than copying its bytes.

Private viewing and preview lifecycle

The API resolves a document through its chain, stage and document IDs. Images and PDFs are streamed with no-store, nosniff and authorization-aware headers. Original download URLs expire after five minutes; anyone holding a valid signed URL can use it until expiry, so it should be treated as a temporary secret.

Office preview fetches the private original, hashes it and caches the PDF at eudr-previews/{hash}/v1.pdf. In-flight requests for the same source share a job. There are at most two in-flight preview jobs per server process; this is not a distributed queue. Viewing or converting evidence does not approve it or change its audit status.

Conversion and browser rendering

Office conversion uses a fresh temporary work directory and LibreOffice profile. The production path requires Linux bubblewrap and prlimit isolation, disables macros and link updates, restricts filesystem and network access, and applies a 60-second conversion timeout. Input and output are bounded at 50 MiB. DOCX/XLSX active-content entries are rejected, and temporary files are removed in finally.

Spreadsheet PDF export includes worksheets with single-page-per-sheet rendering. The React viewer renders PDFs with pdfjs-dist; Vite emits the worker through its worker asset pipeline instead of serving the original .mjs file. The viewer manages thumbnails, page navigation, zoom, keyboard controls, downloads and resource cleanup.

Security boundaries and limits

Router-level role checks protect all EUDR endpoints. created_by is audit metadata, not a per-user ownership filter. The router has no additional brand-specific access scope. File type checks are not antivirus scanning; only forest images receive full image decoding at upload. Converted previews may alter layout, so the original remains authoritative.

Maintenance guide and operational limits

Where to read or change the implementation

Area	Repository location
Database schemas	backend/models/eudrChain.model.js backend/models/eudrParty.model.js backend/models/eudrAllocation.model.js
Routes and request handling	backend/routers/eudrChain.routes.js backend/controllers/eudrChain.controller.js
Workflow and live grouping	backend/services/eudrChain.service.js backend/helpers/timberCft.js
Atomic allocation	backend/services/eudrAllocation.service.js
Saved identities and evidence	backend/services/eudrDirectory.service.js backend/services/eudrFiles.service.js
Storage and conversion	backend/services/wasabiStorage.service.js backend/services/eudrPreview.service.js backend/services/convertXlsxToPDF.service.js
Frontend page and forms	client/OMS/src/pages/EudrTimber.jsx client/OMS/src/components/eudr/StageEditor.jsx client/OMS/src/components/eudr/SavedDetails.jsx
Review and rendering	client/OMS/src/components/eudr/EudrChainView.jsx client/OMS/src/components/eudr/DocumentWorkspace.jsx client/OMS/src/components/eudr/pdfRuntime.js client/OMS/src/utils/eudrDocuments.js

Operational requirements

Use a MongoDB replica set or transaction-capable cluster. On the first indexed write, allocation setup creates required indexes and removes only the obsolete unique container/vendor index. Keep Wasabi objects private and configure storage credentials and bucket settings. Office preview additionally requires the Linux conversion sandbox and an available LibreOffice executable; absence fails closed rather than using an unsafe fallback.

Regression coverage

Backend test files cover chain gating, forest data, invoices, sale links, directory snapshots, duplicate evidence, allocation, previews, PIS and shipment CFT. Separate eudrAllocation.integration.js exercises a temporary replica set; eudrOffice.integration.js uses real conversion fixtures. eudrChain.browser.js and frontend eudrDocuments.test.js cover the viewer and workflow interfaces. Run backend and frontend npm test plus the production build when changing behaviour.

Current limits and safe extension points

Missing-CFT exclusions can make partial timber appear fully covered, and received-volume differences are non-blocking. Invoice uniqueness applies to current invoice_keys, not a permanent registry of all superseded invoice history; the same active invoice cannot be split across separate chains. List and directory results are capped at 50 without pagination. Audit arrays grow inside the chain and have no archival policy here.

Legacy approvals remain until affected stages are edited. Version 2 receiver upgrades archive old evidence and link the prior seller rather than inventing missing buyer data. Old purchase-model EUDR data is not migrated. There is no AI/OCR verification or automatic compliance determination. A future verifier can use verification_method, but must preserve the same validation, role checks, ordered gating and audit rules.